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Data Engineer

PROFESSIONAL SUMMARY:

- With over 4+ years of experience with specialization in data engineering, analytics, and development across various domains.
- Hands-on programmer with extensive experience in Python programming for data processing and automation tasks.
- Expert in designing and managing end-to-end data pipelines, utilizing technologies to enhance data flow and quality.
- Proficient in leveraging Google Big Query for advanced data analytics and reporting in financial and healthcare sectors.
- Skilled in developing and optimizing complex SQL queries for Google Big Query, ensuring optimal performance and scalability.
- Designed and implemented data models for faster query execution and improved system performance.
- Highly proficient in SQL for complex data management, query optimization, and interacting with different database systems.
- Proficient in handling big data technologies such as Apache Beam and Apache Spark for large-scale data processing tasks.
- Utilized Apache Beam for creating unified data processing pipelines that support both batch and stream processing.
- Employed Apache Spark for optimizing large-scale data processing tasks, ensuring high performance and scalability.
- Integrated AWS Glue for seamless data integration and ETL processes, enhancing data quality and consistency.
- Extensive experience in real-time data streaming and processing using Apache Kafka in diverse application scenarios.
- Expert in orchestrating complex workflows using Apache Airflow, enhancing data pipeline reliability and efficiency.
- Managed and optimized large datasets using Google Big Query, enabling efficient data manipulation and retrieval.
- Experienced in managing SQL and No-SQL databases, ensuring data integrity and reliability.
- Well-versed in cloud computing, specifically AWS and Google Cloud, for developing scalable and robust data solutions.
- Proficient in using Docker for application containerization, ensuring consistent environments across development and production.
- Strong background in Agile methodologies, enabling flexible and efficient project management in data engineering projects.
- Experienced in Jenkins and Terraform for continuous integration and effective infrastructure management in data projects.
- Effective communicator in regular stakeholder engagement, ensuring project success and alignment with business goals.

- Skilled in implementing fault-tolerant and highly available data processing systems using Apache Spark and Apache Beam.
- Experienced in optimizing Spark jobs for low latency and high throughput, leveraging advanced techniques such as dynamic resource allocation and partitioning.
- Adept at leveraging Apache Beam's rich set of transforms and IO connectors to build complex data processing pipelines.
- Proficient in using Spark's Data Frame API for efficient data manipulation and analysis, enabling streamlined data workflows.
- Demonstrated ability to integrate Apache Beam pipelines with cloud data warehouses like Google Big Query for scalable and efficient data storage and retrieval.
- Experienced in configuring and managing Spark clusters in both on-premises and cloud environments, ensuring high availability and resilience.
- Strong background in developing data processing applications with Apache Beam's portable runner architecture, allowing cross-platform execution.
- Knowledgeable in utilizing Spark's machine learning libraries for predictive analytics and data-driven decision-making processes.
- Skilled in leveraging Beam's unified programming model to handle both batch and streaming data within a single pipeline, simplifying the development process.

TECHNICAL SKILLS:

Programming Languages:	Python, R, PL/SQL
Data Visualization Tools:	Tableau
Database Technologies:	Microsoft SQL Server, MongoDB, AWS Redshift
Big Data Technologies:	Apache Kafka, Hadoop YARN, Apache Beam, Apache Hive
Cloud Services:	AWS (S3, EMR, Redshift), Google Cloud Storage
Machine Learning:	TensorFlow
ETL Tools:	Informatica PowerCenter, SSIS, Talend
Other Technologies:	Docker, Git, Jenkins, Terraform, Apache Airflow, Databricks, Snowflake
Methodologies:	Agile, Business Analytics, Documentation, CI/CD

PROFESSIONAL EXPERIENCE:

Client: Ericsson

Role: Data Engineering Intern

Duration: Aug 2023 – Dec 2023

Roles & Responsibilities:

- Assisted in the design and implementation of data processing pipelines using Apache Beam and Apache Spark, gaining hands-on experience in large-scale data processing tasks.

- Supported the development and optimization of ETL processes using AWS Glue and Apache Kafka, ensuring efficient data integration and transformation.
- Contributed to the management of data lakes and the optimization of data workflows for enhanced performance and data integrity.
- Assisted in the automation of complex data workflows using Apache Airflow, improving operational efficiency and reliability.
- Participated in the development and optimization of complex SQL queries, enhancing data manipulation and retrieval capabilities.
- Gained experience with cloud technologies such as AWS S3, Redshift, and EMR, supporting data storage and processing initiatives.
- Assisted in implementing monitoring solutions using Dynatrace and Splunk to ensure system performance and reliability.
- Supported DevOps practices by working with Jenkins, GitHub, JFrog, and uDeploy, contributing to continuous integration/delivery pipelines.
- Collaborated with cross-functional teams to support data engineering projects, gaining insights into best practices and industry standards.
- Assisted in developing and testing Apache Beam pipelines for data cleansing and enrichment, improving data quality for downstream analytics.
- Contributed to the implementation of Spark MLlib for machine learning tasks, enabling predictive analytics and enhancing decision-making processes.
- Supported the migration of legacy batch processing jobs to Apache Spark, improving the efficiency and scalability of data workflows.
- Assisted in configuring and managing Spark clusters on AWS EMR, ensuring optimal resource allocation and performance.
- Participated in debugging and troubleshooting Apache Beam pipelines, identifying and resolving issues to maintain data processing reliability.
- Maintained high standards of data security and compliance, assisting in the implementation of data protection measures.

Environment: AWS (S3, Redshift, EMR), Apache Beam, Apache Spark, AWS Glue, Apache Kafka, Apache Airflow, Dynatrace, Splunk, Jenkins, GitHub, JFrog, uDeploy, MongoDB, SQL.

Client: Ericsson

Duration: Sept 2019 - July 2022

Role: Data Engineer

Roles & Responsibilities:

- Leveraged AWS S3 for secure and scalable data storage solutions, addressing key insurance data management needs.
- Designed and implemented data processing pipelines using big data technologies such as Apache Beam and Apache Spark, optimizing Spark jobs for performance and scalability.
- Leveraged AWS S3 for secure and scalable data storage solutions, and AWS Redshift for efficient data warehousing, enhancing data analytics capabilities.
- Utilized Apache Beam for creating unified data processing pipelines that support both batch and stream processing.
- Orchestrated ETL processes using MongoDB, AWS Glue, and Apache Kafka for efficient data processing and management.
- Employed AWS EMR for managing and processing large-scale data, improving data handling capabilities.

- Integrated Apache Airflow for automating complex data workflows, enhancing operational efficiency and data reliability.
- Managed data lakes and optimized data workflows for enhanced performance and data integrity.
- Developed and optimized complex SQL queries for efficient data manipulation and retrieval.
- Implemented monitoring solutions using Dynatrace and Splunk to ensure system performance and reliability.
- Leveraged Jenkins, GitHub, JFrog, and uDeploy for effective DevOps practices and continuous integration/delivery pipelines.
- Provided technical expertise and leadership in data engineering projects, driving innovation and improving data-driven decision-making processes.
- Implemented data aggregation and transformation processes using Apache Spark, reducing data processing time by 30%.
- Designed and developed custom data connectors for Apache Beam pipelines to integrate with various data sources, enhancing data accessibility and flow.
- Utilized Apache Beam's windowing and triggering mechanisms to handle late-arriving data and ensure data consistency in streaming applications.
- Developed Spark Structured Streaming applications to process real-time data streams, improving the responsiveness of data-driven applications.
- Conducted performance tuning and optimization of Spark jobs, including memory management, caching strategies, and parallel processing.
- Ensured compliance with financial regulations and implemented robust data security measures, maintaining high standards of data integrity and protection.

Environment: AWS services (S3, Redshift, EMR, Schema Conversion Tool), MongoDB, Apache Kafka, Apache Airflow, Hadoop YARN, Python, SQL, Big Query, SSIS, compliance with financial regulations, data security, data quality, Shell Scripting, Azure, Azure Data Factory.

Client: HCL Technologies

Duration: Jan 2019 to Sept 2019

Role: Data Analyst (Intern)

Roles & Responsibilities:

- Using SQL queries, extracted, altered, and analyzed relational database data for reporting. Used reports and visualizations to share results with stakeholders. Found improvement possibilities by analyzing data trends, patterns, and anomalies.
- Developed and optimized data processing workflows using Apache Beam and Apache Spark.
- Gained hands-on experience with Hadoop and Apache Flink for distributed data processing tasks.
- Employed Spark for data cleansing, transformation, and analysis, ensuring efficient data processing.
- Utilized Apache Beam for building robust and flexible data pipelines supporting diverse data sources and sinks.
- Applied monitoring tools to track system performance and detect anomalies, ensuring data pipeline reliability.
- Engaged in ETL (Extract, Transform, Load) process design and development to import data from diverse sources into destination databases or data warehouses. Used Apache Airflow, Talend, or AWS Glue to write ETL scripts. Verified and tested ETL data for quality and integrity.
- Learned and used Spark for big data analytics. Used Scala or Python to create Spark apps for data cleansing, transformation, and analysis. Optimized Spark tasks for performance and scalability via partitioning and caching.

- Got familiar with Kafka for real-time communications and streaming. Became familiar with Kafka architecture, producers, consumers, and subjects. Added Kafka to data pipelines for streaming processing.
- Learned Amazon S3, Redshift, Glue, Athena, EMR, Kinesis, and Lambda for data analytics and processing. AWS data workload deployment and management best practices. Tried AWS tools and services in laboratories, exercises, and projects.
- Attended SQL, ETL, Spark, Kafka, and AWS training, workshops, and online courses. Requested mentorship from senior team members or managers to expedite learning. Explored data analytics and processing technology, tools, and methods.
- In all work and interactions, maintained professionalism, promptness, and reliability. Followed corporate policies and conduct. Teamwork and morale improved when team members and stakeholders collaborated well. Clear, polite communication requested clarity when needed and updated progress and problems. Developed a growth mentality, accepted comments, and strove for development.

Environment: SQL, ETL, Spark, Kafka, AWS (S3, Redshift, Glue, Athena, EMR, Kinesis, Lambda).

Education:

- Bachelor of Technology (B.Tech) from BhojReddy Institute of Technology for Women , Hyderabad, Telangana, India. – 2020
- Master's in professional science, Data Science from UMBC, Maryland - 2024